

6. Complex Power in AC Circuits

Introduction to Electric Circuits Supplemental Instruction

Cooper McAllister

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This document is not a substitute for a textbook or classroom instruction. It may contain errors and is intended to be used only to the extent that it is helpful to supplement students' learning experience.

1 Three Phase Line and Phase Voltages and Currents

Three Phase circuits can be connected either in either wye (Y) or delta (Δ) configuration. The relationship between line voltages and phase voltages, as well as between line currents and phase currents, can be shown to be as follows. These equations apply for magnitudes or RMS values.

In Y configuration:

$$V_l = \sqrt{3}V_p$$

$$I_l = I_p$$

In Δ configuration:

$$V_l = V_p$$

$$I_l = \sqrt{3}I_p$$

When looking at a circuit diagram, it is easy to see for each configuration which quantities are the same and which are different.

References

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- T. L. Floyd and D. M. Buchla, Principles of Electric Circuits: Conventional Current, 10th ed. Hoboken, NJ: Pearson Education, 2020.
- O. Ortiz. 2025. ENGR 2053 Intro to electric circuits [PowerPoint slides].